

Mental Health Data Integration & Analytics

Understanding how mental health affects patients' lives, treatments, and healthcare costs

Our project focuses on three main aspects:

- Behavior
- Treatment
- Cost

Integrating multiple data sources to support better healthcare decisions

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Lab 1: Project Initialization & Data Profiling

MENTAL HEALTH

- Stress & Anxiety
- Sleep Patterns
- 10,000 records



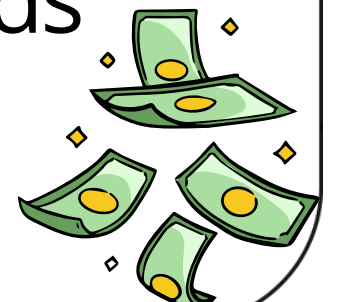
TREATMENT

- Diagnosis
- Therapy & Medication
- 500 records



INSURANCE

- Costs & Charges
- BMI & Region
- 1,338 records



No missing values (0%) • Some duplicates exist • Structured data (CSV)

Integrating multiple data sources for comprehensive analysis

Lab 2: Structured vs. Unstructured Data

Structured Data

- Organized in tables
- Easy to analyze
- Example: Insurance, Treatment

Unstructured Data

- Text-based data
- Difficult to process
- Example: Patient narratives

Data Transformation

- Clean text (remove noise)
- Extract key information (disease, cost)
- Identify categories (insurance, treatment)
- Convert into structured table

Transforming raw text into structured data for analysis

Lab 3: Data preprocessing & Cleaning for Integration

Mental Health Data

- Inconsistent date formats
- Gender variations
- Country name issues

Treatment Data

- Duplicate patient IDs
- Date format issues
- Invalid severity values

Insurance Data

- Invalid age values
- Inconsistent gender
- Outliers in charges

Cleaning Actions :

Standardize formats • Remove duplicates • Fix missing/invalid values • Normalize categories

Improving data quality before integration

Lab 4: Schema Matching

- Source A: Mental Health Diagnosis & Source B: Medical Insurance

- Summary :
- Match: 2 attributes
 - Partial Match: 0 attributes
 - No Match: 15 attributes

Source A Attribute	Source B Attribute	Match Type	Confidence
Age	Age	✓ Match	High
Gender	Sex	✓ Match	High
Patient ID	-	✗ No Match	Low
Diagnosis	-	✗ No Match	Low
Symptom Severity	-	✗ No Match	Low
Mood Score	-	✗ No Match	Low
Therapy Type	-	✗ No Match	Low

Lab 5: Schema Mappings

- Table 1: Schema Mapping Tablebetween Source A and Source B

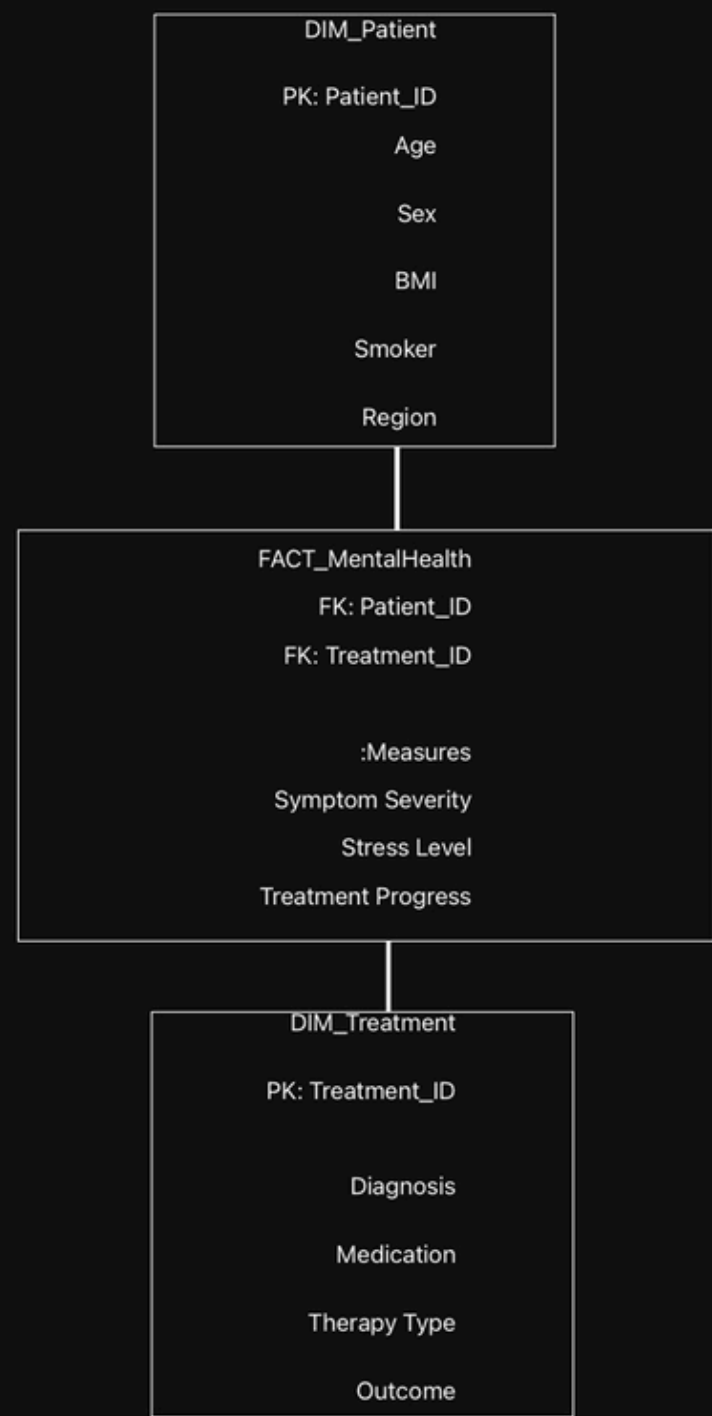
Source Attribute	Target Attribute	Mapping Type	Short Description
Age	Age	1:1	Same attribute in both datasets
Gender	Sex	1:1	Same concept - different naming
Patient ID	-	Conditional	Internal identifier only in Source A

- Table 2: mapping rules.

Rule	Rule Expression	Explanation
1	If Age exists → map directly to Age	same attribute in both datasets
2	If Gender = Male/Female → map to Sex	same meaning, different name

Lab 6: Data Warehouse design & Data Marts

1. Star Schema



2. Dimension tables

Table name	Row inserted (Number)	Source used	Key Attributes
DIM_Patient	10	Medical Insurance Dataset	Age, Sex, BMI, Smoker, Region
DIM_Treatment	10	Mental Health Diagnosis Dataset	Diagnosis, Medication, Therapy, Outcome
FACT_MentalHealth	10	All The 3 Source	Severity, Stress, Progress measures

Lab 7: Data Warehouse design & Data Marts

Strategic Questions based on the dashboards:

Which diagnosis has the highest average stress level?

Major Depressive Disorder has the highest average stress level.

Which therapy type shows the highest treatment progress?

Mindfulness-Based Therapy shows the highest treatment progress.

Which region has the highest average BMI?

The Southeast region has the highest average BMI

What is the relationship between depression level and treatment progress?

Patients with low depression levels show better treatment progress

How does anxiety relate to treatment progress?

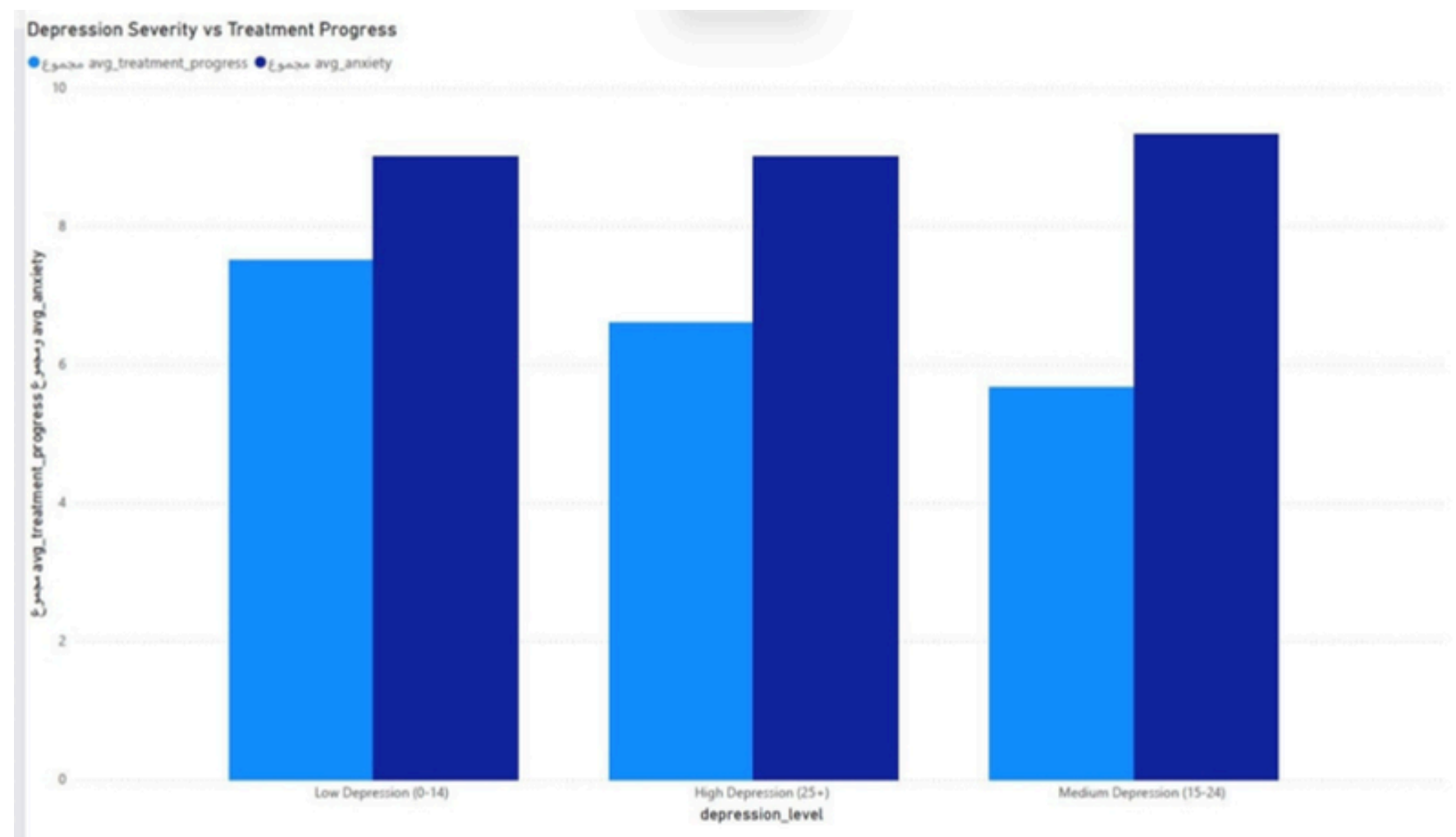
Higher anxiety levels are associated with lower treatment progress

OLAP Analysis:

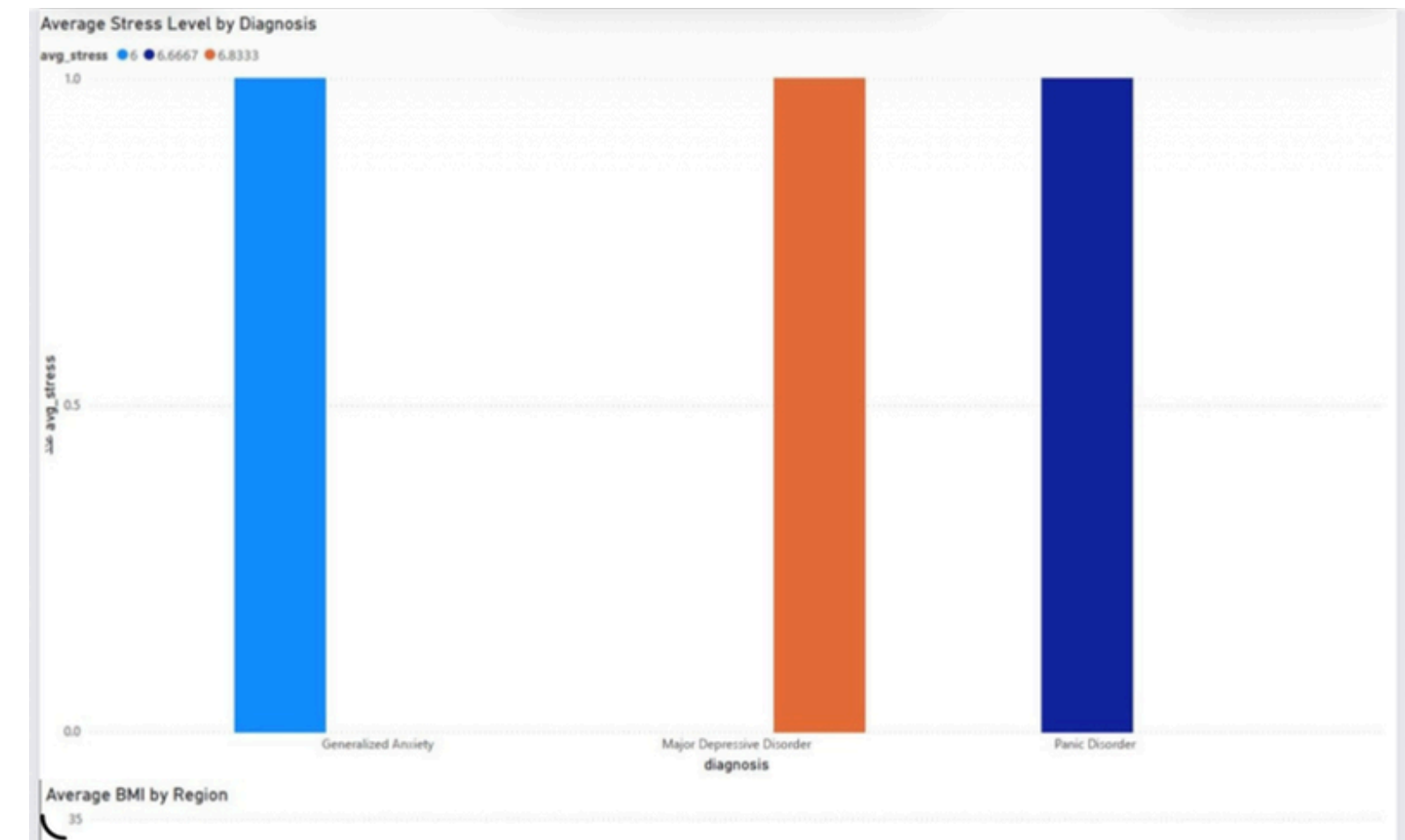
We used OLAP analysis to view the data from different dimensions, such as diagnosis, therapy .type, region, depression level, and anxiety level

The charts helped us compare average stress level, treatment progress, and BMI. This allowed us to answer the strategic questions and understand which factors have the strongest effect on patient recovery

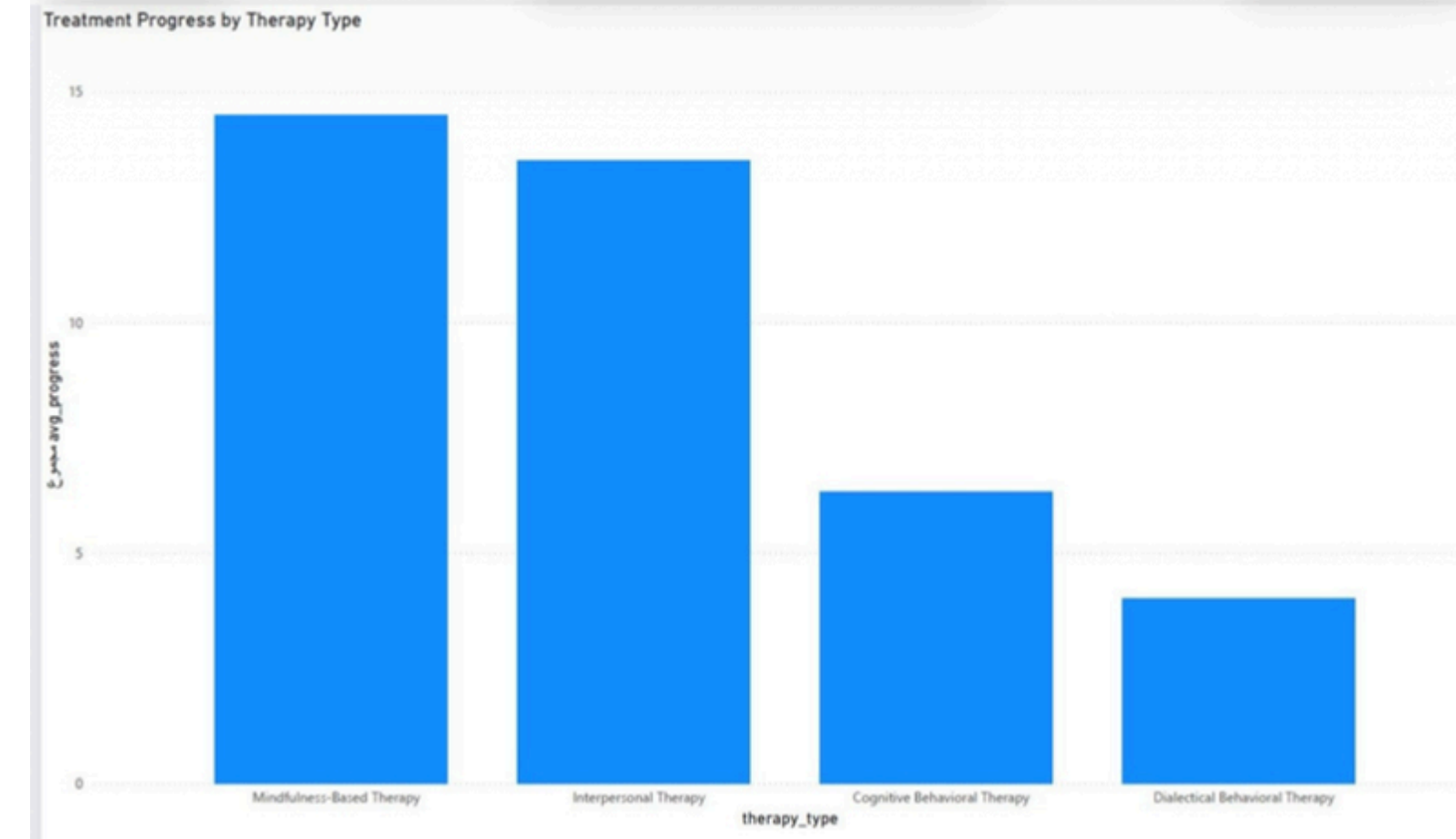
Key Results & Dashboard .



Low depression levels achieved better treatment progress, while anxiety negatively affects progress



Major Depressive Disorder had the highest average stress level •



Mindfulness-Based Therapy showed the best treatment progress

Final Recommendations:

- .Improve stress management programs for patients with high stress levels
- .Use Mindfulness-Based Therapy more frequently because it showed the best treatment progress
- Focus on early intervention for depression cases to improve recovery outcomes
- .
- Monitor anxiety levels during treatment because anxiety may negatively affect treatment progress
- .
- .Use data analysis to support better healthcare decisions

Lessons Learned:

- Data analysis helped us identify patterns between mental health factors and treatment progress
- .
- Comparing diagnosis, stress level, and treatment progress showed that depression and anxiety can affect recovery outcomes
- .
- Using dashboards made it easier to understand the results and compare patients by diagnosis, therapy type, and depression level
- .
- OLAP analysis helped us view the data from different dimensions and answer the strategic questions clearly
- .
- .Clean and integrated data is important because it gives more accurate and reliable results